VIDYA ACADEMY OF SCIENCE AND TECHNOLOGY

PCCSL507 – NETWORKS LAB

EXPERIMENT 13

Implement RIPv2 Routing

1 Aim

To configure RIPv2 (Routing Information Protocol Version 2) on the routers and verify connectivity using the routing table and ping commands.

2 Requirements

- Cisco Packet Tracer
- Four Cisco 2811 routers
- One PC
- Copper straight-through cable
- Serial DCE/DTE cables

3 Theory

RIPv2 is a dynamic routing protocol used by routers to exchange routing information and automatically learn routes to remote networks. RIP uses hop count as its routing metric.

RIPv2 is enabled using the `version 2` command. The `no auto-summary` command disables automatic classful route summarization. The `network` command specifies the networks on which RIP operates.

Routes learned through RIP are identified by the letter `R` in the routing table. RIP has an administrative distance of 120.

4 Network Topology

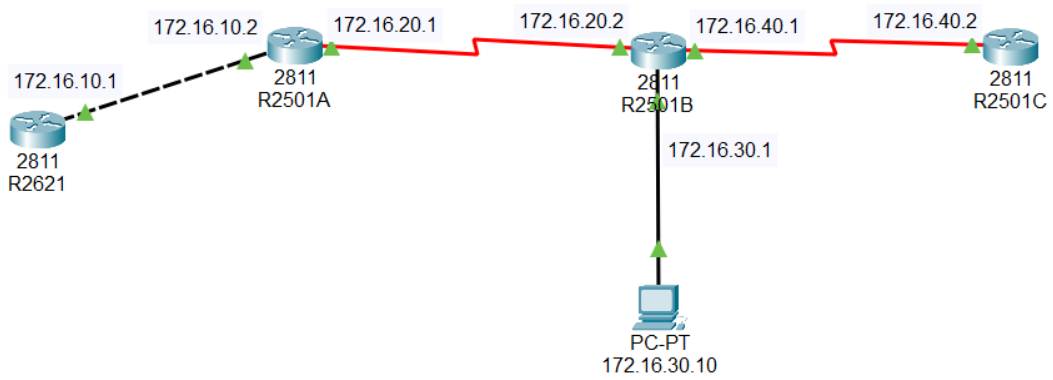


Figure 1: RIPv2 Network Topology

5 IP Address Allocation

Device	Interface	IP Address	Subnet Mask
R2621	FastEthernet0/0	172.16.10.1	255.255.255.0
R2501A	FastEthernet0/0	172.16.10.2	255.255.255.0
R2501A	Serial0/0/0	172.16.20.1	255.255.255.0
R2501B	Serial0/0/0	172.16.20.2	255.255.255.0
R2501B	FastEthernet0/0	172.16.30.1	255.255.255.0
R2501B	Serial0/0/1	172.16.40.1	255.255.255.0
R2501C	Serial0/0/0	172.16.40.2	255.255.255.0
PC	FastEthernet0	172.16.30.10	255.255.255.0

Table 1: IP Address Allocation

5.1 Network Addresses

Network	Network Address
R2621 – R2501A	172.16.10.0/24
R2501A – R2501B	172.16.20.0/24
R2501B – PC	172.16.30.0/24
R2501B – R2501C	172.16.40.0/24

Table 2: Network Addresses

6 Connectivity

Device	Interface	Connected Device	Interface
R2621	FastEthernet0/0	R2501A	FastEthernet0/0
R2501A	Serial0/0/0 (DCE)	R2501B	Serial0/0/0 (DTE)
R2501B	Serial0/0/1 (DCE)	R2501C	Serial0/0/0 (DTE)
R2501B	FastEthernet0/0	PC	FastEthernet0

Table 3: Network Connectivity

7 Procedure

1. Connect the routers and PC according to the given topology.
2. Assign IP addresses to all router interfaces.
3. Configure the DCE interfaces with the required clock rate.
4. Enable all required router interfaces using `no shutdown`.
5. Configure the IP address, subnet mask, and default gateway on the PC.
6. Verify the interface status using `show ip interface brief`.
7. Configure RIPv2 on all routers.
8. Verify the RIP configuration using `show ip protocols`.
9. Display the routing table using `show ip route`.
10. Verify connectivity using the `ping` command.

8 Configuration

8.1 Router R2621

```
enable
configure terminal

interface fastethernet0/0
ip address 172.16.10.1 255.255.255.0
no shutdown
exit

end
```

8.2 Router R2501A

```
enable
configure terminal

interface fastethernet0/0
ip address 172.16.10.2 255.255.255.0
no shutdown
exit

interface serial0/0/0
ip address 172.16.20.1 255.255.255.0
clock rate 64000
no shutdown
exit

end
```

8.3 Router R2501B

```
enable
configure terminal

interface serial0/0/0
ip address 172.16.20.2 255.255.255.0
no shutdown
exit

interface fastethernet0/0
ip address 172.16.30.1 255.255.255.0
no shutdown
exit

interface serial0/0/1
ip address 172.16.40.1 255.255.255.0
clock rate 64000
no shutdown
exit

end
```

8.4 Router R2501C

```
enable
configure terminal

interface serial0/0/0
ip address 172.16.40.2 255.255.255.0
no shutdown
```

```
exit

end
```

9 PC Configuration

Configure the PC using the following values:

```
IP Address      : 172.16.30.10
Subnet Mask     : 255.255.255.0
Default Gateway : 172.16.30.1
```

10 RIPv2 Configuration

10.1 Router R2621

```
enable
configure terminal

router rip
version 2
no auto-summary
network 172.16.10.0

end
```

10.2 Router R2501A

```
enable
configure terminal

router rip
version 2
no auto-summary
network 172.16.10.0
network 172.16.20.0

end
```

10.3 Router R2501B

```
enable
configure terminal

router rip
version 2
no auto-summary
network 172.16.20.0
```

```
network 172.16.30.0
network 172.16.40.0

end
```

10.4 Router R2501C

```
enable
configure terminal

router rip
version 2
no auto-summary
network 172.16.40.0

end
```

11 Verification

11.1 Interface Verification

On each router, verify the interface status using:

```
show ip interface brief
```

All connected interfaces should be in the `up/up` state.

11.2 RIP Verification

Verify the RIP configuration using:

```
show ip protocols
```

Display the routing table using:

```
show ip route
```

RIP-learned routes are indicated by the letter `R`.
For example, R2621 should learn:

```
R 172.16.20.0/24
R 172.16.30.0/24
R 172.16.40.0/24
```

11.3 Connectivity Verification

From R2621:

```
ping 172.16.30.1
ping 172.16.40.2
```

From R2501C:

```
ping 172.16.10.1
```

From PC:

```
ping 172.16.30.1
ping 172.16.10.1
ping 172.16.40.2
```

Successful replies confirm that RIPv2 routing and end-to-end connectivity are working correctly.

12 Result

RIPv2 was successfully configured on all routers. The routers dynamically learned the remote networks through RIPv2, and connectivity between the networks was successfully verified using the routing table and ping commands.